

1A: <Harsith902>

My Enigma number

E =

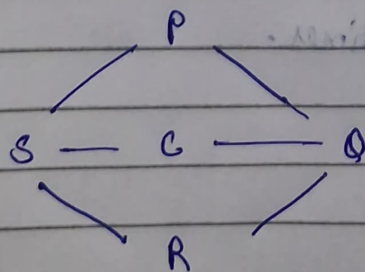
github username = harsith902

position	character	value	multiplier	
1	h	8	1	8
2	a	1	2	2
3	r	18	4	72
4	s	19	8	152
5	i	9	16	144
6	+	20	32	640
7	h	8	64	512
8	9	9	128	1152
9	0	0	256	0
10	2	2	512	1024
				<u>3706</u>

E = 3706

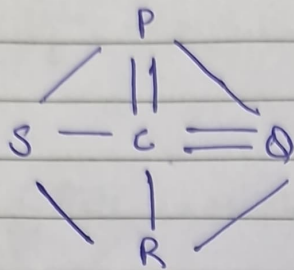
* level - 1A The Bridges of city.

Initial map

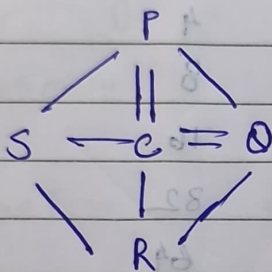


$3 \Rightarrow P-C \Rightarrow$ odd no.s
 $7 \Rightarrow Q-C$ so 2 bridges
 $0 \Rightarrow R-C$ form next
 $6 \Rightarrow S-C$ to them.

Final Bridge



2. Draw Bridges



Total = 4 + 2 + 2 + 1 + 1
 Bridges (outer island) [double Bridges] single bridge to center

R. My guess is walk is possible.

3. Attempt 1: X X

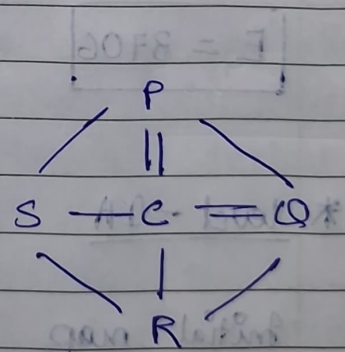
① $S \rightarrow P \rightarrow \boxed{C \rightarrow R} \rightarrow S$

$\boxed{R \leftarrow C} \leftarrow Q \leftarrow P \leftarrow C$

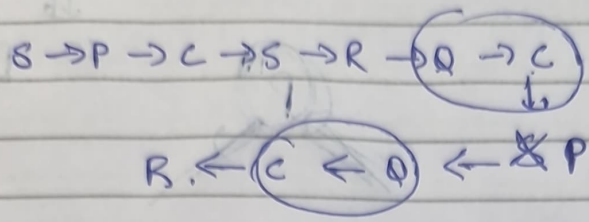
↓
Q.

⇒ This path is not possible.

as $C \rightarrow R$ is used 2 times.

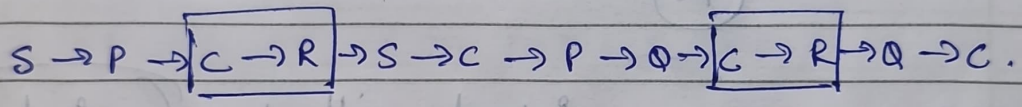
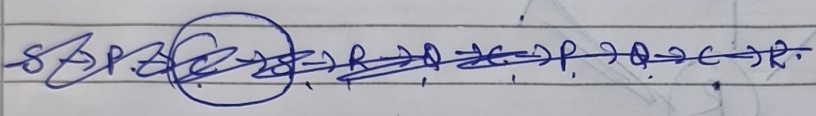


② Attempt 2. ✓✓



Here the double bridges are used for $Q \rightarrow C$ path.
and the walk is possible.

③ Attempt 3. X X.



This path is not possible as $C-R$ is used 2 times.

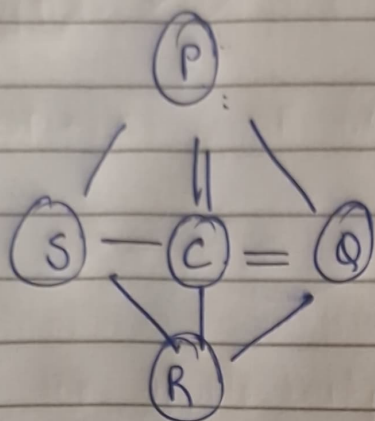
4. Count

Island	The Bridges
P	4 Bridges
Q	4
R	3
S	3
C	6

P, Q, C $\rightarrow$ Even counts.

(S, R) $\rightarrow$ odd counts, _/_/_

Que 8.



① so all the even counts will be paired and will be used for (arrival-departure)

② hence ~~that~~ if we use P, Q, C as the endpoints we will not be able to complete walk, ~~cause it would~~ cause we cannot use one bridge multiple times

Eg: $P \rightarrow Q \rightarrow R \rightarrow C \rightarrow P$

$Q \leftarrow C \leftarrow S$
now ($Q \rightarrow R$ and $Q \rightarrow P$
Bridges are used)

③ using S, R as endpoints will create a ~~way~~ as walk as they will have an odd bridge that could complete the walk.